

HANK YAN

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SUMMARY OF QUALIFICATIONS

- Foundational experience with Python and PyTorch through projects in LoRA implementation and large language model fine-tuning
- Experience using Claude Code to review AI-generated educational labels and check data quality across 248 annotations and 26 concepts
- Built a minimal LoRA layer in PyTorch and learned how to freeze model weights, train adapter parameters, and merge the adapter into the base layer
- Interested in software development, machine learning, and using AI tools to solve technical problems

TECHNICAL SKILLS

- **Programming:** Python
- **AI and Machine Learning:** PyTorch, LoRA, parameter-efficient fine-tuning, transformer fundamentals, LLM inference
- **Data:** JSON, data preparation, data validation, annotation review
- **Tools:** GitHub, Claude Code, LM Studio, PyCharm, Conda

RELEVANT EXPERIENCE

AI Education Data Verification Intern

August 2026 – September 2026

Remote

- Used Claude Code to find and list 248 low-confidence labels across 26 educational concepts
- Manually checked each label based on the related concept and found 6 incorrect labels
- Recorded the 6 corrections in a Word report; all proposed changes were accepted and used in the revised dataset

TECHNICAL PROJECTS

LoRA Implementation and LLM Fine-Tuning

February 2026 - Present

Python, PyTorch, Qwen 3.5-4B, LM Studio

- Implemented a minimal LoRA layer in PyTorch by freezing the base weights and training two smaller low-rank matrices
- Checked that the base weights stayed unchanged during training and compared the number of trainable parameters between full fine-tuning and LoRA
- Merged the trained LoRA adapter into the base layer and checked that the merged and unmerged models produced the same output
- Used an AI-assisted workflow to fine-tune Qwen 3.5-4B on dialogue data from subtitle files
- Processed subtitle files into JSON, manually checked the training data, managed training checkpoints, and ran the model locally in LM Studio
- Replaced an unreliable OCR method with script-based subtitle extraction to create more consistent training data

EDUCATION

Bachelor of Applied Science – Computer Engineering

September 2026 - Present

University of Waterloo, Waterloo, ON

- Relevant Courses: Communication in the Engineering Profession, Fundamentals of Programming, Classical Mechanics, Engineering Profession and Practice, Project Studio, Linear Algebra for Engineering, Calculus 1 for Engineering
- Awards: International Student Entrance Scholarship, University of Waterloo President's Scholarship (2026)